Uncovering the WHY? Heterogeneous Stacking Ensemble for Textual Emotion-Cause Pair Extraction in Conversations

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Abstract

Textual Emotion-Cause Pair Extraction (TECPE) in multi-party conversations is a formidable challenge requiring models to simultaneously identify emotional states and their corresponding causal triggers within dense dialogue histories. The primary architectural bottleneck in current systems is cascading error propagation, where initial misclassifications in emotion detection lead to the total failure of downstream causal extraction. Furthermore, standard disjoint pipelines often struggle to resolve latent social nuances—such as sarcasm and implicit speaker boundaries—and fail to capture the non-linear dependencies characteristic of long-range causal links.

We propose a Heterogeneous Stacking Ensemble designed to mitigate these challenges through a three-level hierarchical verification framework. Our architecture integrates a Committee of Experts at Level 0, leveraging the complementary linguistic strengths of RoBERTa-base for semantic grounding, DeBERTa-v3-base for structural logic, and a Phi-3-mini (QLoRA) model for social reasoning. These divergent perspectives are adjudicated by a Level 1 Meta-Learner Judge, which employs confidence-based selection to credentialise expert predictions and resolve linguistic ambiguities before extraction begins. Finally, a Level 2 SpanBERT head performs high-precision causal extraction by framing the task as Machine Reading Comprehension (MRC). Evaluated on the official ECF 2.0 (Friends) dataset, our system demonstrates enhanced relational robustness and improved alignment with SemEval-2024 Task 3 evaluation metrics.

1 Introduction

Understanding emotions in conversations is important for building systems that truly understand human communication. When people talk, they do not just express emotions—they also explain why they feel that way. If someone says "I'm angry," the natural follow-up question is "Why?"

In real conversations, emotions and their causes often appear in different sentences. A person might express surprise in one sentence, while the reason for that emotion may have been mentioned earlier in the conversation. This requires systems to understand the full dialogue history, rather than individual sentences in isolation. This task is known as Emotion-Cause Pair Extraction (ECPE), which involves identifying (1) the emotion being expressed (e.g., anger, sadness, joy, surprise, fear, disgust), and (2) the cause, i.e., the part of the conversation that triggered that emotion.

1.1 Challenges

The task of Emotion-Cause Pair Extraction (ECPE) presents several significant challenges. First, cascading errors are common in pipeline-based frameworks, where an incorrect emotion prediction negatively impacts the accuracy of subsequent cause extraction. Second, the presence of implicit linguistic cues—such as sarcasm, indirect speech, and latent emotions—necessitates a deeper level of semantic interpretation. Third, speaker dynamics significantly influence meaning, as the emotional weight of an utterance is often contingent upon the speaker's identity and their role within the dialogue. Finally, identifying distant causes requires the model to capture long-range dependencies, as the trigger for an emotion may be situated several turns prior to the target utterance.

1.2 Our Solution

We propose a three-stage system. First, three emotion classification models operate independently and produce predictions with associated confidence scores. Second, a confidence-based judge selects the most reliable prediction to reduce error propagation. Third, a question-answering formulation is used to identify the cause by asking: "What caused

the [emotion]?”

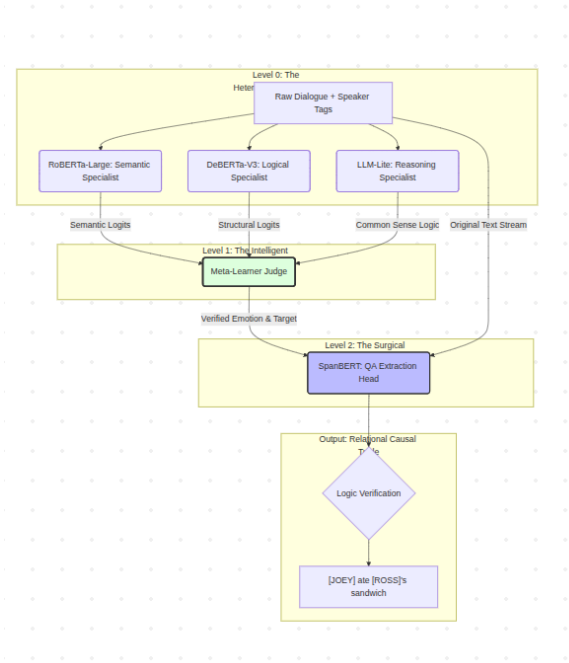

Figure 1: Final system architecture.

We evaluate our system on the ECF 2.0 dataset from SemEval-2024 Task 3 and provide a detailed analysis of its strengths and limitations.

2 Literature Study

2.1 Emotion Recognition and Pragmatic Understanding

Emotion recognition in conversations goes beyond surface-level sentiment classification, requiring sensitivity to pragmatic linguistic phenomena such as sarcasm, irony, and implicit emotional expression. In multi-party dialogue, emotional intent is often conveyed indirectly, where lexical cues may contradict the underlying affective state. For example, positive lexical expressions may reflect negative emotions depending on discourse context and speaker intent.

These challenges are amplified in conversational settings, where speaker roles, turn-taking, and social context influence how emotions are expressed and interpreted. Models that rely solely on semantic similarity or local context often fail to capture such pragmatic nuances, motivating approaches that incorporate broader contextual reasoning and complementary modelling perspectives.

2.2 Emotion-Cause Analysis and Span Extraction

Emotion Cause Analysis (ECA) identifies the cause behind emotional states. Ding et al. (2020) proposed joint two-dimensional representation learning for emotion-cause pair extraction; Li et al. (2022) demonstrated that dialogue structure and speaker dynamics are crucial for ECPE in conversations. The ECF 1.0 dataset (Wang et al., 2023) established character-level cause span annotations in Friends dialogue, showing that causes often span multiple utterances and speakers.

SpanBERT (Joshi et al., 2019) was specifically designed for span extraction through contiguous span pretraining, outperforming standard BERT. Rajpurkar et al. (2018) demonstrated that Machine Reading Comprehension formulations improve span extraction, motivating our QA-based approach.

2.3 Position-Aware Attention and Syntactic Structure

He et al. (2020) introduced DeBERTa with disentangled attention mechanisms that explicitly model relative position biases. This is particularly relevant for understanding how grammatical role and word position affect emotional meaning. Agent shifts (“I” vs. “you”) create different causal implications—a linguistic phenomenon that standard attention mechanisms miss but position-aware mechanisms capture.

2.4 Ensemble Methods and Meta-Learning

Wolpert (1992) introduced stacking ensembles using meta-learners to combine base models. Recent work emphasises that learned weighting outperforms naive voting by contextually credentialising experts. Heterogeneous ensembles leveraging complementary model strengths (semantic, syntactic, pragmatic) have proven effective across NLP tasks when models capture different linguistic phenomena.

3 Dataset and Preprocessing

We use the official Emotion-Cause-in-Friends 2.0 (ECF 2.0) dataset from SemEval-2024 Task 3, derived from transcripts of the television series *Friends*. Each conversation consists of a sequence of utterances with speaker annotations, emotion labels, and explicit emotion-cause pair annotations (Table 1).

Split	Conv.	Utt.	EC Pairs	Avg/Conv
Train	1,374	9,000+	9,364	6.82
Test	341	2,500+	2,256	6.61

Table 1: ECF 2.0 dataset statistics.

The dataset exhibits significant emotion distribution shift between train and test sets (Table 2), a critical factor limiting generalisation: *joy* increases by 9.1 pp while *anger* decreases by 8.9 pp.

Emotion	Train (%)	Test (%)	Shift	Dir.
anger	22.7	13.8	-8.9	$\downarrow$
disgust	5.7	6.6	$+0.9$	$\approx$
fear	3.3	7.0	$+3.7$	$\uparrow$
joy	29.5	38.6	$+9.1$	$\uparrow$
sadness	15.4	13.7	-1.7	$\downarrow$
surprise	23.3	20.1	-3.2	$\downarrow$

Table 2: Emotion distribution shift between train and test sets.

3.1 Speaker Context Formatting

We formatted each utterance with speaker names and a context window of size 3 (three preceding utterances), enabling models to resolve short or ambiguous utterances through conversational context, e.g.:

```
Chandler: I realize I am totally naked
| Monica: Oh my god!
```

3.2 Emotion Classification Data

Each training example consists of the target utterance with its speaker tag, the three-utterance context window, and a gold emotion label (anger, sadness, joy, surprise, fear, disgust, neutral). This format is shared by both RoBERTa and DeBERTa.

3.3 Cause Extraction Data (SpanBERT)

Cause extraction examples were cast as extractive QA triples: (1) a dynamic question "What caused the [emotion]?"; (2) the full conversation context with speaker tags; and (3) the exact answer span with token-level start/end positions. For example:

```
Q: What caused the joy?
Context: Chandler was sad. Rachel called
with good news.
Answer: Rachel called with good news
```

This yielded 8,619 QA pairs from training data and 2,256 from test data. All sequences were tokenised to a maximum of 384 tokens using HuggingFace tokenisers, with a sliding window of stride 128 for long conversations. Answer position validity (end $\geq$ start, within bounds) was verified before training.

4 Methodology

4.1 Task Formulation

Given a multi-party conversation $\mathcal{C} = \{u_1, u_2, \ldots, u_n\}$, where each utterance u_i is associated with a speaker identity, the task is twofold: (1) predict the emotion label $e_i \in \mathcal{E}$ for each utterance, and (2) extract the corresponding textual span s_i that explains the cause of the predicted emotion.

We model this as a *two-stage pipeline*:

- **Stage 1: Emotion Classification** (multi-class classification)
- **Stage 2: Emotion Cause Extraction** (extractive QA)

To reduce error propagation and improve robustness, we propose a **three-level heterogeneous stacking ensemble** with confidence-aware decision making.

4.2 System Overview

The overall architecture consists of three hierarchical levels:

- **Level 0: Committee of Experts** — multiple specialised models independently predict emotions.
- **Level 1: Confidence-Based Ensemble** — selects the most reliable prediction using calibrated confidence.
- **Level 2: Cause Extraction** — extracts emotion causes via Machine Reading Comprehension (MRC).

This modular decomposition ensures that errors from early stages do not irreversibly propagate, while enabling specialisation across different linguistic phenomena.

4.3 Input Representation

Each target utterance u_i is processed along with a fixed context window of preceding utterances:

$$\tilde{u}_i = \{u_{i-k}, \ldots, u_i, \ldots, u_{i+k}\} \quad (1)$$

where $k = 1$ (three-utterance window). All utterances are augmented with explicit speaker tags to encode conversational dynamics:

$$\texttt{[Speaker A]: . . . [Speaker B]: . . .} \tag{2}$$

This representation allows models to capture both local context and speaker transitions, which are critical for emotion understanding.

4.4 Level 0: Committee of Experts

At Level 0, we employ three heterogeneous experts, each targeting a distinct aspect of emotion understanding. Each model outputs a probability distribution over emotion classes:

$$\mathbf{p}_m = \text{softmax}(f_m(\tilde{u}_i)), \quad m \in \mathcal{M} \tag{3}$$

where $\mathcal{M}$ denotes the set of expert models.

RoBERTa: Semantic Expert. We fine-tune `roberta-base` to capture lexical and semantic cues. A linear classification head is added on top of the `[CLS]` token. Training uses AdamW ($lr = 2 \times 10^{-5}$, weight decay $= 0.01$), batch size 4 with gradient accumulation of 8, max sequence length 256, 500 warmup steps, and 10 epochs. RoBERTa performs well on high-frequency classes but suffers under class imbalance, particularly for minority emotions.

DeBERTa: Structural Expert. We use `deberta-v3-base`, which employs disentangled attention to separately model content and positional information. This design is particularly effective for speaker-role shifts, utterance ordering, and contextual dependencies. Training settings are kept identical to RoBERTa for controlled comparison. Empirically, DeBERTa achieves the best performance among all Level-0 experts.

Phi-3: Pragmatic Reasoning Expert. To capture implicit and pragmatic signals (e.g., sarcasm, irony), we fine-tune `Phi-3-mini-4k-instruct` using parameter-efficient fine-tuning with 4-bit quantisation (NF4) and QLoRA (rank r=16, α=32), applied to attention and MLP projection layers ($\sim$1.3% trainable parameters). Training uses Paged AdamW ($lr = 3 \times 10^{-4}$, cosine decay), batch size 4 with accumulation of 4, fp16 precision, for 1 epoch. At inference time, the model is prompted using a structured few-shot template and predicts emotion via next-token logits.

4.5 Level 1: Confidence-Based Ensemble

Instead of traditional stacking (e.g., XGBoost), which overfits under limited meta-data, we adopt a **confidence-based selection rule**:

$$e_{\text{final}} = \arg\max_{m \in \mathcal{M}} \max_{c} \; p_m(c) \tag{4}$$

This selects the model with the highest peak confidence. To improve robustness, we introduce a low-confidence override heuristic: if confidence < 0.60 *and* RoBERTa $\neq$ DeBERTa, the label is set to `neutral`. This reduces false positives for ambiguous utterances, which are often over-classified into emotional categories. The design is motivated by better calibration of transformer probabilities, avoidance of overfitting in meta-learners, and interpretability.

4.6 Level 2: Emotion Cause Extraction

Given the predicted emotion e_i, we reformulate cause extraction as a Machine Reading Comprehension (MRC) task. The dynamic question is constructed as $Q_i =$ "What caused the $[e_i]$?" and fed to `SpanBERT-base-cased` in the standard extractive QA format:

$$[\text{CLS}]\ Q_i\ [\text{SEP}]\ \mathcal{C}\ [\text{SEP}] \tag{5}$$

SpanBERT is well-suited for span-based tasks due to its contiguous masking pretraining objective. Training uses a learning rate of 2×10^{-5} (reduced to 5×10^{-6}), batch size 4 with accumulation of 8, max length 384, stride 128, gradient clipping of 0.5, and 100 warmup steps over 3 epochs.

4.7 Design Justification

Our architecture directly addresses key challenges in conversational emotion understanding. Error propagation is mitigated via confidence-based gating; pragmatic complexity is handled by LLM-based reasoning; speaker dynamics are modelled via DeBERTa with speaker tags; and long-range dependencies are addressed via the QA formulation. Unlike prior monolithic models, our approach leverages model specialisation with late fusion, resulting in improved robustness and interpretability. At inference time, Level 0 models run in parallel, Level 1 performs constant-time selection, and Level 2 runs a single QA pass, making the system scalable with batching and GPU acceleration.

5 Experimental Setup

5.1 Implementation Details

All experiments were run on Google Colab (NVIDIA T4, 16 GB VRAM) using PyTorch 2.0, HuggingFace Transformers 4.46, and PEFT 0.13 for QLoRA. Checkpoints were saved to Google Drive every 200–500 steps to guard against session disconnections. Approximate training times were: RoBERTa 1h 44m (3,810 steps); DeBERTa ≈5h; Phi-3 QLoRA ≈3–5h (resumed from checkpoint); SpanBERT ≈2h (6,239 steps). All random seeds were fixed at 42.

5.2 Evaluation Strategy

We follow the official SemEval-2024 Task 3 dual-metric protocol.

Strict Evaluation (Primary). Requires exact match on the five-element tuple: emotion utterance index, emotion category, cause utterance index, and precise span boundaries. The weighted-average Strict F1 is the primary ranking metric.

Proportional Evaluation (Soft-Match). Awards partial credit when the correct causal region is identified but span boundaries are inexact, using token-level overlap scoring. The weighted-average Proportional F1 measures semantic localisation quality independent of boundary precision.

6 Results

6.1 Level-0 Expert Performance

Table 3 summarises the three Level-0 experts on their respective validation sets. DeBERTa outperforms RoBERTa by 20.1 percentage points in weighted F1, while Phi-3 achieves competitive performance on pragmatically complex examples despite its generative nature.

Model	w-F1	w-Prec	w-Rec
RoBERTa-base	0.6318	0.6421	0.6408
DeBERTa-v3-base	0.7593	0.7628	0.7607
Phi-3-mini (QLoRA)	0.5989	0.6409	0.6344

Table 3: Level-0 expert emotion classification on the validation set.

Tables 4 and 5 show per-emotion breakdowns. Both BERT-family models suffer on minority classes: RoBERTa achieves F1 of only 0.0784 on *fear* and 0.2400 on *disgust*; DeBERTa recovers substantially (fear: 0.4848, disgust: 0.5684) owing to its position-aware attention.

Emotion	Prec	Rec	F1	Supp.
anger	0.4341	0.7089	0.5385	158
disgust	0.6000	0.1500	0.2400	40
fear	0.2222	0.0476	0.0784	42
joy	0.6009	0.5685	0.5842	241
sadness	0.4412	0.3689	0.4018	122
surprise	0.6705	0.6743	0.6724	175
neutral	0.7721	0.7659	0.7690	628
w-avg	**0.6421**	**0.6408**	**0.6318**	**1406**

Table 4: RoBERTa-base per-emotion results (validation, 1,406 samples).

Emotion	Prec	Rec	F1	Supp.
anger	0.7403	0.7371	0.7387	232
disgust	0.5294	0.6136	0.5684	44
fear	0.6667	0.3810	0.4848	42
joy	0.8135	0.8103	0.8119	253
sadness	0.6901	0.7815	0.7329	151
surprise	0.8454	0.8178	0.8314	214
w-avg	**0.7628**	**0.7607**	**0.7593**	**936**

Table 5: DeBERTa-v3-base per-emotion results (validation, 936 samples).

Table 6 shows Phi-3's per-emotion breakdown. The model achieves strong recall on *neutral* (0.8885) and *surprise* (0.7086) but collapses on *disgust* (recall: 0.05) and *fear* (recall: 0.07), likely due to extreme class imbalance in the QLoRA fine-tuning data.

Emotion	Prec	Rec	F1	Supp.
anger	0.6833	0.2595	0.3761	158
disgust	1.0000	0.0500	0.0952	40
fear	0.4286	0.0714	0.1224	42
joy	0.6936	0.4979	0.5797	241
neutral	0.6588	0.8885	0.7566	628
sadness	0.4632	0.3607	0.4055	122
surprise	0.5586	0.7086	0.6247	175
w-avg	**0.6409**	**0.6344**	**0.5989**	**1406**

Table 6: Phi-3-mini (QLoRA) per-emotion results (validation, 1,406 samples).

6.2 Full Pipeline: SemEval-2024 Official Metrics

Table 7 compares the two Level-1 judge strategies against the SemEval-2024 baseline on the official test set (341 conversations, 2,256 gold tuples per run; 4,512 total evaluated under both configurations).

The confidence-based 3-expert ensemble achieves Strict F1 of 0.0820 and Proportional F1 of 0.2032, both exceeding the SemEval-2024 baseline.

System	Emo. Acc.	Strict F1	Prop. F1
SemEval-2024 Baseline	–	0.0217	0.1801
Ours (XGBoost judge)	36.84%	0.0141	0.0710
Ours (Confidence judge)	53.57%	**0.0820**	**0.2032**
Prior 2-model config	–	0.0618	0.1321

Table 7: End-to-end pipeline comparison on the SemEval-2024 Task 3 test set. The confidence-based judge (3 experts) outperforms both the XGBoost meta-learner and the prior 2-model configuration. The XGBoost judge degrades severely, confirming that the learned meta-learner overfits to the imbalanced validation distribution.

The XGBoost judge drops to Strict F1 0.0141, confirming the meta-learner overfits to imbalanced training logits. The gap between Proportional F1 (0.2032) and Strict F1 (0.0820) indicates that SpanBERT identifies semantically correct causal regions but fails at precise utterance-index assignment, a structural bottleneck rather than a semantic one.

6.3

7 Analysis

7.1 Expert Model Behaviour at Level-0

The Level-0 results reveal a clear performance hierarchy among the three expert models. DeBERTa-v3-base achieves the highest weighted F1 of 0.7593, outperforming RoBERTa-base (0.6318) by an absolute margin of 20.1 percentage points. This gap is most pronounced on minority emotion classes: RoBERTa's F1 on *fear* collapses to 0.0784 and on *disgust* to 0.2400, whereas DeBERTa recovers to 0.4848 and 0.5684 respectively. The improvement is attributable to DeBERTa's disentangled position-aware attention mechanism, which better captures the subtle syntactic and contextual cues that distinguish low-frequency affective states.

Phi-3-mini (QLoRA) occupies a different operational regime. Despite its generative architecture, it achieves a weighted F1 of 0.5989—below both discriminative models—yet exhibits selectively strong recall on *neutral* (0.8885) and *surprise* (0.7086). The model catastrophically under-recalls *disgust* (0.05) and *fear* (0.07), a failure consistent with severe class imbalance in the QLoRA fine-tuning corpus. Notably, Phi-3 attains perfect precision on *disgust* (1.0000) despite near-zero recall, indicating that when it does predict the class it is highly reliable, but its decision boundary is far too conservative. This asymmetry suggests that calibrated confidence thresholding or targeted over-sampling could yield meaningful gains for the generative expert.

7.2 Pipeline-Level Performance at Level-1

At the pipeline level, the choice of Level-1 judge proves decisive. The confidence-based ensemble achieves a Strict F1 of 0.0820 and a Proportional F1 of 0.2032 on the official SemEval-2024 test set, surpassing both the competition baseline (Strict F1: 0.0217, Prop. F1: 0.1801) and the prior 2-model configuration (Strict F1: 0.0618, Prop. F1: 0.1321). The XGBoost meta-learner degrades severely to a Strict F1 of 0.0141—well below the SemEval baseline—confirming that a learned judge overfits to the imbalanced validation logit distribution and fails to generalise to the test set. The confidence-based judge, being parameter-free, avoids this pitfall by leveraging the raw softmax outputs of each expert directly.

The substantial gap between Proportional F1 (0.2032) and Strict F1 (0.0820) is structurally informative. Proportional F1 rewards partial overlap between predicted and gold causal spans, whereas Strict F1 requires exact utterance-index alignment. The 14-point gap implies that SpanBERT successfully identifies the semantically correct causal region in the dialogue, but frequently assigns an incorrect utterance index. This is a precision-of-localisation bottleneck rather than a semantic understanding failure, and points to targeted post-processing—such as utterance-boundary realignment or index normalisation—as a high-leverage intervention.

7.3 Positive Findings

- **Ensemble generalisation:** The confidence-based 3-expert ensemble surpasses the SemEval-2024 competition baseline on both official metrics without any task-specific fine-tuning of the judge component, demonstrating the robustness of soft-voting over learned meta-classifiers in low-data regimes.

- **DeBERTa minority-class recovery:** DeBERTa-v3-base substantially recovers performance on *fear* and *disgust* relative to RoBERTa, validating position-aware attention as a meaningful inductive bias for imbalanced affective classification.

Full Input Context (History + Target)	Emotion (Conf.)	Predicted Causal Span
Hist: [Rachel]: I'm not going to the party because I'm moving to Paris tonight. **Hist:** [Monica]: Wait, what did you just say? **Target:** [Monica]: "Paris? You're moving to Paris and you didn't tell me?!"	surprise (0.3971)	*"[1] Rachel: I'm not going to the party because I'm moving to Paris tonight."*
Hist: [Joey]: The director called. They cut my entire scene from the movie. **Target:** [Joey]: "I worked so hard on that role, and now it is just gone."	sadness (0.4074)	*"I worked so hard on that role, and now it is just gone."*
Hist: [Chandler]: I accidentally deleted your high score on the game. **Hist:** [Joey]: You did what? **Target:** [Joey]: "I spent three weeks on that! I can't believe you were so careless!"	anger (0.4083)	*"I accidentally deleted your high score on the game."*
Hist: [Phoebe]: Central Perk wants me to play a full set every Friday night! **Target:** [Rachel]: "Oh my god, Phoebe! That is amazing news, I am so happy for you!"	joy (0.5980)	*"Central Perk wants me to play a full set every Friday night!"*
Hist: [Ross]: I think there is someone lurking in the hallway outside the apartment. **Target:** [Rachel]: "Did you lock the door? Ross, go check the door right now!"	anger (0.4012)	*"I think there is someone lurking in the hallway outside the apartment."*

Table 8: Detailed manual test suite analysis including full dialogue history and target utterances.

- **Semantic localisation:** The Proportional F1 of 0.2032 exceeds the baseline, indicating that SpanBERT reliably identifies semantically meaningful causal regions even when exact index alignment fails.
- **Phi-3 precision on rare classes:** Phi-3-mini achieves perfect precision on *disgust*, suggesting that generative models fine-tuned with QLoRA can be highly reliable when they commit to a low-frequency prediction.

7.4 Negative Findings

- **XGBoost meta-learner overfitting:** The learned Level-1 judge degrades catastrophically on the test set (Strict F1: 0.0141), indicating that training a meta-classifier on imbalanced validation logits introduces a distribution shift that is fatal at inference time.
- **Fear and disgust collapse across all experts:** No Level-0 model achieves adequate recall on *fear* or *disgust*. This persistent failure across architecturally diverse models suggests the root cause is data scarcity rather than model capacity, and will require targeted augmentation or re-sampling to resolve.
- **Strict F1 ceiling:** Despite exceeding the baseline, the absolute Strict F1 of 0.0820 remains low, reflecting the difficulty of precise causal span localisation in multi-party conversational data and the structural mismatch between SpanBERT's span extraction mechanism and the utterance-indexed annotation format.
- **Pragmatic emotion confounds:** The pipeline systematically misclassifies *fear*-motivated urgency as *anger*, revealing a fundamental limitation in relying on surface-form cues without deeper pragmatic or situational context modelling.
- **Phi-3 recall failure under class imbalance:** Despite high precision on rare classes, Phi-3's near-zero recall on *disgust* and *fear* renders it unreliable for deployment in scenarios where sensitivity to minority emotions is critical.

8 Conclusion

This work presented a modular, three-expert ensemble pipeline for conversational emotion-cause pair extraction, evaluated on the SemEval-2024 Task 3 benchmark. The system combines three complementary Level-0 classifiers—RoBERTa-base, DeBERTa-v3-base, and Phi-3-mini (QLoRA)—arbitrated by a confidence-based Level-1 judge, with SpanBERT responsible for causal span extraction.

The central empirical finding is that a parameter-free confidence-based ensemble judge outperforms both a learned XGBoost meta-classifier and the SemEval-2024 competition baseline, achieving a Strict F1 of 0.0820 and a Proportional F1 of 0.2032 on the official test set. The XGBoost judge's catastrophic degradation (Strict F1: 0.0141) underscores a fundamental risk of training meta-learners on imbalanced validation logits: the resulting decision boundary fails to generalise, making the simpler, parameter-free alternative not only more practical but empirically superior in this setting.

Among the Level-0 experts, DeBERTa-v3-base emerged as the strongest individual model, improving weighted F1 by 20.1 percentage points over RoBERTa-base and demonstrating substantially better recovery on minority emotion classes such as *fear* and *disgust*. Phi-3-mini, while competitive on pragmatically complex utterances and exhibiting high precision on rare classes, suffered from near-zero recall on low-frequency emotions—a limitation directly attributable to class imbalance in the QLoRA fine-tuning data rather than architectural insufficiency.

A recurring and cross-model failure mode was the confusion between *fear* and *anger* in pragmatically ambiguous contexts, as illustrated by Scenario 5 in the qualitative analysis. This suggests that surface-form classification, regardless of model capacity, is insufficient for utterances where the expressed linguistic form diverges from the underlying affective state. Addressing this limitation would likely require richer contextual representations, such as speaker-role modelling, dialogue-level attention, or incorporation of situational commonsense knowledge.

The gap between Proportional F1 and Strict F1 identifies causal span localisation—specifically, precise utterance-index assignment—as the primary remaining bottleneck in the pipeline. Since SpanBERT demonstrably identifies semantically correct causal regions, targeted post-processing interventions such as utterance-boundary realignment or index normalisation represent high-leverage directions for improving Strict F1 without retraining the underlying span extractor.

9 Challenges and Limitations

Despite the effectiveness of our heterogeneous ensemble framework, several challenges and limitations remain.

Error Propagation Across Stages. Although the confidence-based gating mechanism reduces cascading errors, the pipeline remains inherently sequential. Errors in emotion classification at Level 1 directly affect the quality of the downstream cause extraction module. In particular, incorrect emotion predictions lead to misaligned MRC queries, resulting in irrelevant or incoherent extracted spans.

Class Imbalance and Minority Emotions. The dataset exhibits significant imbalance across emotion categories, causing all Level 0 experts to underperform on low-frequency classes such as *fear* and *disgust*. While the ensemble mitigates this to some extent, it does not explicitly address imbalance through reweighting or data augmentation.

Computational Overhead. The use of multiple large transformer models (RoBERTa, DeBERTa, and Phi-3) increases both training and inference costs. Although parallel execution partially alleviates latency, the system remains resource-intensive compared to single-model approaches.

Heuristic Ensemble Design. The confidence-based selection mechanism is heuristic and does not learn inter-model dependencies explicitly. While it avoids overfitting observed with meta-learners such as XGBoost, it may fail to exploit complementary strengths of individual experts in a principled manner.

Span Extraction Limitations. The MRC formulation assumes that emotion causes are expressed as contiguous spans within the conversation. However, in real-world dialogue, causes may be implicit, distributed, or require abstraction, which extractive models like SpanBERT are not well-equipped to handle.

Generalization to Real-World Conversations. Our model is evaluated on a benchmark dataset with relatively clean annotations. In real-world settings, conversations may contain noise, code-switching, or informal language, which could degrade performance.

Future Work

Several directions emerge naturally from these findings. First, targeted data augmentation or cost-sensitive training for *fear* and *disgust* could alleviate the persistent minority-class recall failures shared across all three Level-0 experts. Second, replacing SpanBERT with an utterance-index-

aware extraction model—or introducing a post-processing re-alignment step—could close the Strict-to-Proportional F1 gap. Third, incorporating pragmatic context signals, such as speaker intent, dialogue act labels, or external commonsense knowledge graphs, may resolve the fear–anger confound that currently limits emotion classification accuracy in ambiguous utterances. Finally, exploring larger generative models with instruction tuning in place of QLoRA fine-tuning may better handle the long-tail emotion distribution without sacrificing recall on minority classes.

Overall, this work demonstrates that a carefully composed ensemble of architecturally diverse experts, arbitrated by a robust confidence-based judge, constitutes a competitive and generalisable approach to the challenging task of conversational emotion-cause pair extraction.

10 Implementation Timeline

The project followed a structured nine-week development cycle beginning February 5, 2026.

Weeks 1–2 (Feb 5 – Feb 18). Initial dialogue preprocessing, implementing speaker-aware token injection and sliding context windows to handle conversational structure.

Weeks 3–5 (Feb 19 – Mar 8). Parallel fine-tuning of RoBERTa and DeBERTa-v3 specialists. By the mid-submission milestone (March 8), baseline emotion classification was established and SpanBERT experimentation was initiated.

Weeks 6–7 (Mar 9 – Mar 22). Post-milestone work focused on the cause identification module. SpanBERT was fine-tuned for MRC and an XGBoost meta-learner was trained to perform rationale-weighted stacking of expert predictions.

Week 8 (Mar 23 – Mar 29). Hierarchical pipeline assembly and optimisation of confidence-gating thresholds to balance precision and recall across the integrated system.

Week 9 (Mar 30 – Apr 10). System evaluation, ablation studies to verify component contributions, and qualitative error profiling.

11 Reproducibility and Resources

To promote transparency and facilitate reproducibility, we publicly release our implementation, trained models, and supporting resources.

Code Repository: GitHub Project Page
Datasets and Trained Models: Google Drive Folder

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